Higher National Diploma in Quantity Surveying

2nd Year, 2nd Semester Examination – 2021

QS 22023 – Building Economics II

Instructions for Candidates:
Answer five questions only.
All questions carry equal marks.

No. of questions : 06
No. of pages : 4
Time :Three hours

Question 01

I. What is the purpose of Cost Management? (03 Marks)

II. What are the tools and techniques used for construction cost management? (List out at least four methods) (04 Marks)

III. What are the components in the total building cost? (04 Marks)

IV. Explain the relationship between cost vs. price. (04 Marks)

V. What are the types of cost advice provided within a construction contract? (05 Marks)

(Total 20 Marks)

Question 02

Cost is a major concern over any construction project. Most of the construction projects are prone to risk related cost overruns. Hence Cost plan has been identified as an essential process to mitigate risk related to construction cost overruns.

I. Explain the cost management cycle by relating the stages of a construction project. (06 Marks)

II. What are the key purposes of cost planning? (State at least 4 factors) (04 Marks)

III. Briefly explain the main stages of cost planning. (06 Marks)

IV. What is the difference between a cost estimate and a cost plan? (04 Marks)

(Total 20 Marks)

Question 03

I. What are methods of preliminary estimation? (04 Marks)

II. Imagine that you are a Quantity Surveyor at a High-rise building construction project. The Client of your project asked you to provide an estimate at the initial stage of the project. However, the estimate you prepared was found to be exceeding the Client's budget. What are the possible actions you could take at this situation? (05 Marks)

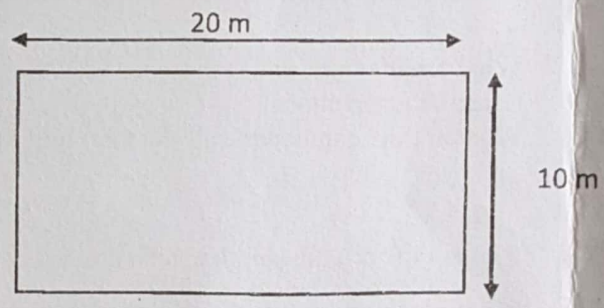
III. Mention the factors considered when selecting a method for preliminary estimation. (03 Marks)

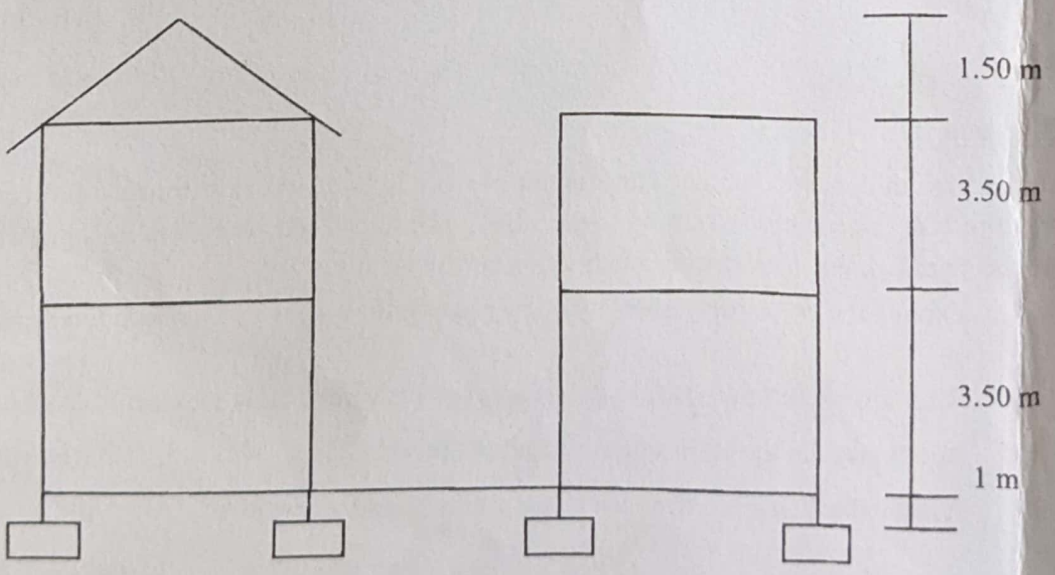
IV. Use the following data to carry out preliminary estimating. (08 Marks)

a) A preliminary estimate for hospital and it needs to be accommodated 1000 patients. You have recently completed a similar project cost data for Rs. 47,000,000.00 and it accommodates 235 patient beds at the same urban location. List out any assumptions.

b) Prepare an estimate for when, the total cost to build a 2-storey villa if the construction cost/m3 of similar construction is Rs. 5, 000, 000.00. You may estimate the cost based on,

- Pitch roof
- Flat roof





(Total 20 Marks)

Question 04

I. Labor, material, plant, overhead and profit are basic elements of unit rate. What are the other considerations that need to be included in the unit rate to provide a more accurate and reliable unit rate? (04 Marks)

II. What are the basic details required to build a labor rate for permanent works? (04 Marks)

III. You have built rate for suspended ceiling works but when prepared to market conditions, it is lower/ higher than the market rate. What are the reasons for the lower/ higher rate? (04 Marks)

IV. What are the factors considered when taking a decision to hire or purchase a plant for a construction company? (04 Marks)

V. You have been appointed as head of the tendering Department in a contracting organization. You need to build an all-in material unit rate for your ongoing tendering projects. What are the things you need to consider? (04 Marks)

(Total 20 Marks)

Question 05

I. What are the design factors influencing the cost of the project? (List out at least five major factors). (05 Marks)

II. What are the costs included in the Life Cycle Cost (LCC)? (03 Marks)

III. Briefly explain the purpose of Life Cycle Costing (LCC). (04 Marks)

IV. List out the major roles of Quantity Surveyor in a construction project? (04 Marks)

V. What are the key skills needed for a Quantity Surveyor? (04 Marks)

(Total 20 Marks)

Question 06

1. What is meant by Earn Value Management/ Earned Value Analysis in construction? Briefly explain. (04 Marks)
2. Briefly explain the following.
   a. Plan Value
   b. Earn Value
   c. Actual Cost (03 Marks)
3. Explain advantages in AOA analysis and Bar Chart analysis for project planning? (02 Marks)
4. Explain Total float, Free Float and Critical path related to AOA analysis? (03 Marks)
5. Find out ES, EF, LS, LF and TF of following activities. (08 Marks)